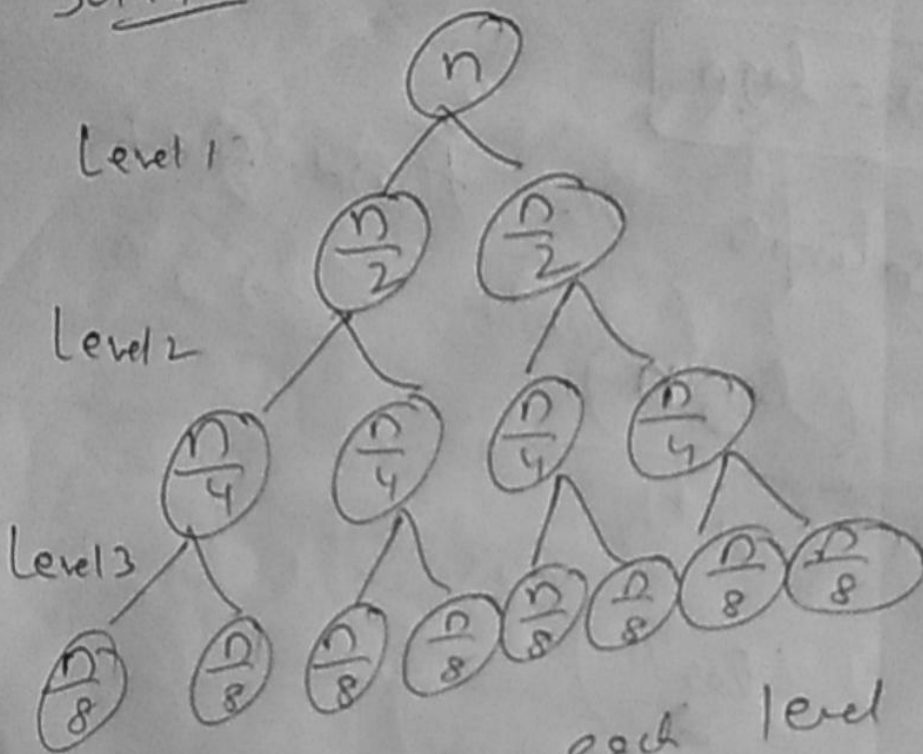


④ Recursive Tree Method

Problem 1

$$T(n) = 2T\left(\frac{n}{2}\right) + cn$$

Solution



The Cost at each level is cn

$$\therefore cn + cn + cn$$
$$= \boxed{3cn}$$

The value 3 depends on half the tree levels.

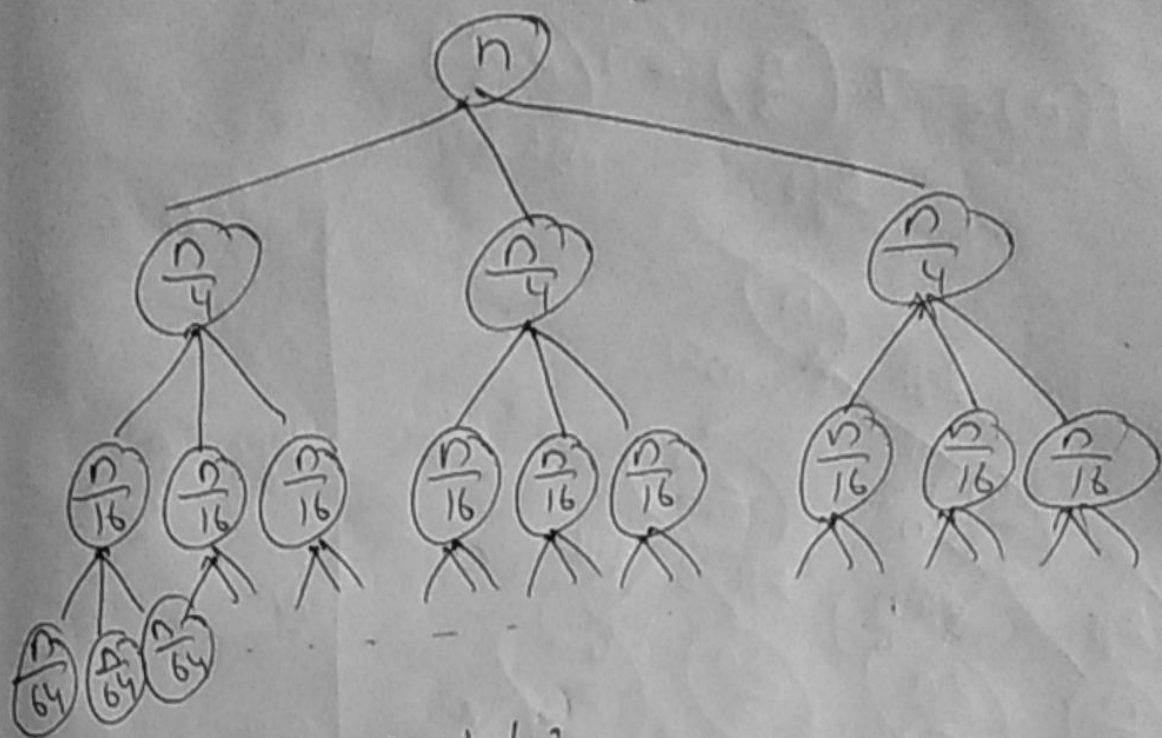
∴ So it represents $\boxed{\log}$ value. Replace 3 with $\log$ value.

$$= \log n \cdot \cancel{c} \cdot n$$

Ignore the constant. So the time complexity is

$$\boxed{O(n \log n)}$$

$$(2) \quad T(n) = 3 T\left(\frac{n}{4}\right) + cn^2$$



Cost calculation

First level $\rightarrow cn^2$

Second level, we have 3 children.

$$\text{So, } c \cdot 3 \left(\frac{n}{4}\right)^2$$

Third level, we have 9 children.

$$\text{So, } c \cdot 9 \left(\frac{n}{16}\right)^2$$

Fourth level, we have 27 children

$$\text{So, } c \cdot 27 \left(\frac{n}{64}\right)^2$$

$$= cn^2 + 3c \left(\frac{n}{4}\right)^2 + 9c \left(\frac{n}{16}\right)^2 + 27c \left(\frac{n}{64}\right)^2 + \dots$$

$$= cn^2 + 3c \frac{n^2}{16} + 9c \frac{n^2}{16^2} + 27c \frac{n^2}{64^2} + \dots$$

$$= cn^2 + cn^2 \left(\frac{3}{16}\right) + cn^2 \frac{9}{16^2} + cn^2 \frac{27}{64^2} + \dots$$

$$= cn^2 + cn^2 \left(\frac{3}{16}\right)^1 + cn^2 \left(\frac{3}{16}\right)^2 + cn^2 \left(\frac{3}{16}\right)^3 + \dots$$

$$= cn^2 \left[1 + \left(\frac{3}{16}\right)^1 + \left(\frac{3}{16}\right)^2 + \left(\frac{3}{16}\right)^3 + \dots \right]$$

It follows Geometric progression
when $r < 1$, the formula is $\boxed{\frac{1}{1-r}}$

$$= cn^2 \left[\frac{1}{1-r} \right] \quad \text{where } r = \frac{3}{16}$$

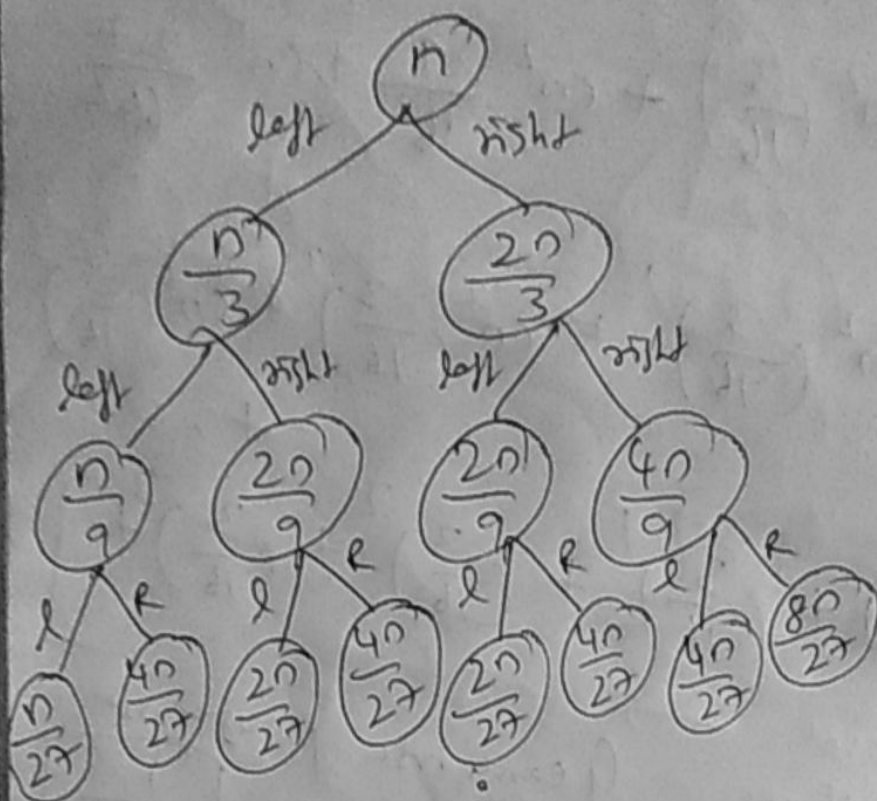
$$= cn^2 \left[\frac{1}{1-\frac{3}{16}} \right]$$

$$= cn^2 \left[\frac{1}{\frac{13}{16}} \right]$$

$$= cn^2 \cdot \frac{16}{13}$$

Ignore the constant
Complexity is
So, the time $\boxed{O(n^2)}$

$$T(n) = T\left(\frac{n}{3}\right) + T\left(2\frac{n}{3}\right) + n$$



Generalise left & right hand

Left	Right
$\frac{n}{3^k} = 1$ $n = 3^k$ Apply log on both sides $\log n = k$ Time Complexity = $n \cdot k$ Maximum $k =$ Minimum $k =$	$\left(\frac{n}{3}\right)^k = 1$ $n = \left(\frac{3}{2}\right)^k$ Apply log on both sides $\log_{3/2} n = k$ Time Complexity = $n \cdot k$ Maximum $k =$ Minimum $k =$

$$\text{left} = \frac{1}{3}$$

$$\text{right} = \frac{2}{3}$$

$$\log t = \frac{1}{3} \times \frac{1}{3} = \frac{2}{9}$$

$$\text{Right} = \frac{n}{3} \times \frac{2}{3} = \frac{25}{9}$$

$$\text{left} = \frac{20}{3} \times \frac{1}{3} = \frac{20}{9}$$

$$\text{figt} = \frac{2n}{3} \times \frac{2}{3} = \frac{4n}{9}$$

$$\text{left} = \frac{n}{9} \times \frac{1}{3} = \frac{n}{27}$$

$$\text{rigel} = \frac{20}{9} \times \frac{2}{3} = \frac{40}{27}$$

$$\text{left} = \frac{2^n}{9} \times \frac{1}{3} = \frac{2^n}{27}$$

$$\text{night} = \frac{20}{9} \times \frac{2}{3} = \frac{40}{27}$$

$$\text{left} = \frac{2n}{9} \times \frac{1}{3} = \frac{2n}{27}$$

$$\text{MSH} = \frac{20}{9} \times \frac{2}{3} = \frac{40}{27}$$

$$\text{left} = \frac{4n}{9} \times \frac{1}{3} = \frac{4n}{27}$$

$$\text{night} = \frac{40}{9} \times \frac{2}{3} = \frac{80}{27}$$